The Intelligent Manufacturing in Civil Engineering

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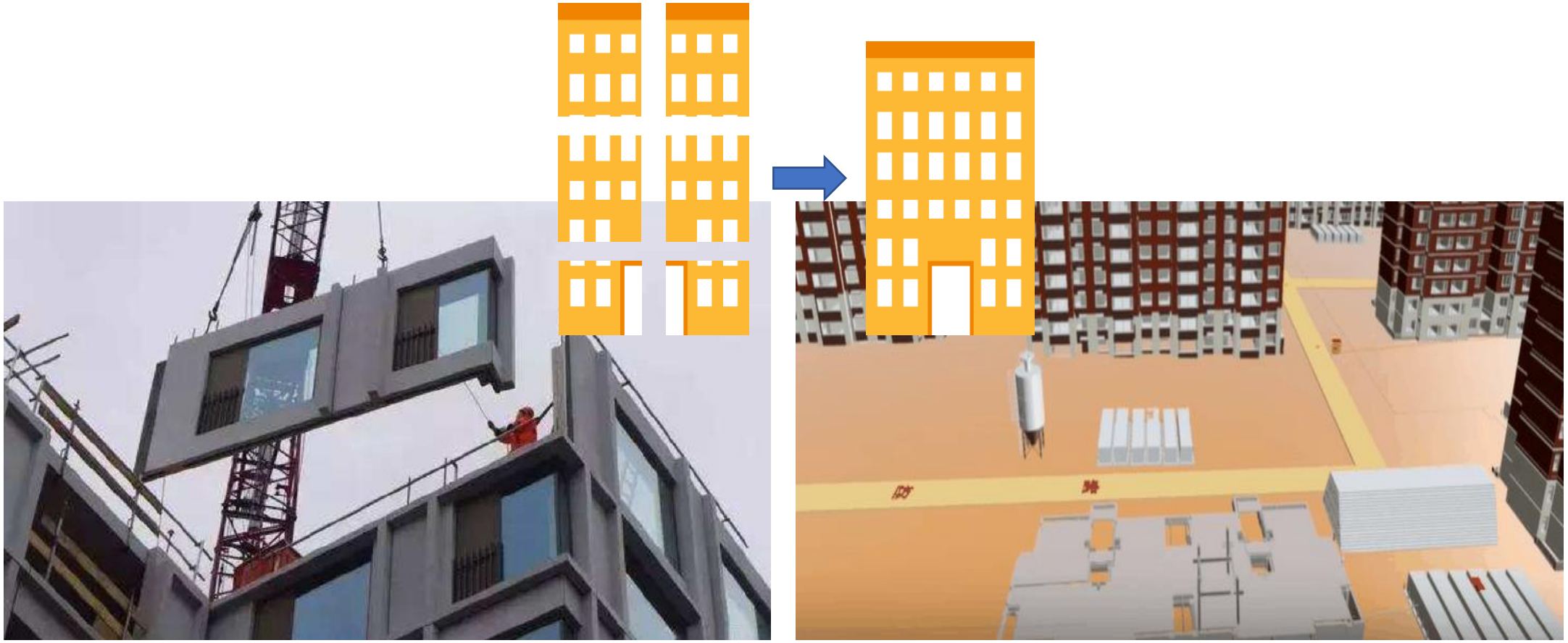
Intelligent Manufacturing

- **Definition:**

  "Intelligent manufacturing" (IM) means using the combined intelligence of people, processes and machines, to upgrade the design, production, management, and integration of the whole life cycle of a product. Its purpose is to optimize manufacturing resources, improve business value and safety, and reduce waste.

- **Key technologies:**
  - IoT
  - Cyber-physical systems (CPSs) ~~~ Digital twin
  - Data storage and cloud computing
  - Big data analytics (BDA)
  - Machine to machine communication

Intelligent Manufacturing of Prefabricated Components(PC)



Intelligent Manufacturing of Prefabricated Components(PC)

- Industrial robot and concrete formwork
- Steel cage and formwork integrated assembly equipment
- Automatic binding or welding equipment for steel cage
- Digital precise distribution technology
- Transportation equipment for finished prefabricated components
- Factory information management system based on BIM

Immovability of production

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Product → ? → Architecture

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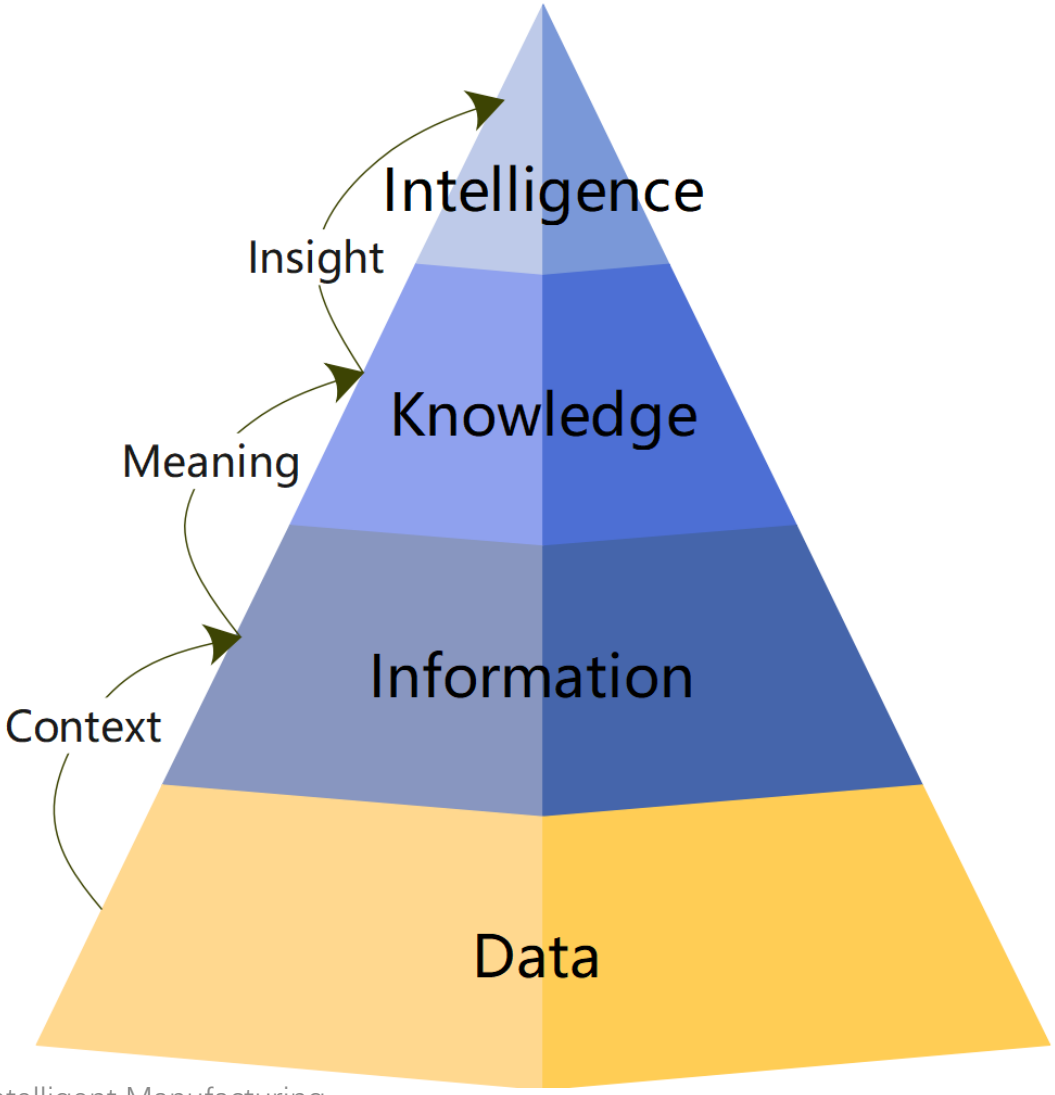
Discreteness of manufacturing

Intelligent Manufacturing → Tools / Inspirations / Methods ? → Intelligent Construction

Intelligent Construction

- Intelligence of decision
- Intelligence of knowledge extraction
- Intelligence of information analysis
- Intelligence of data collection

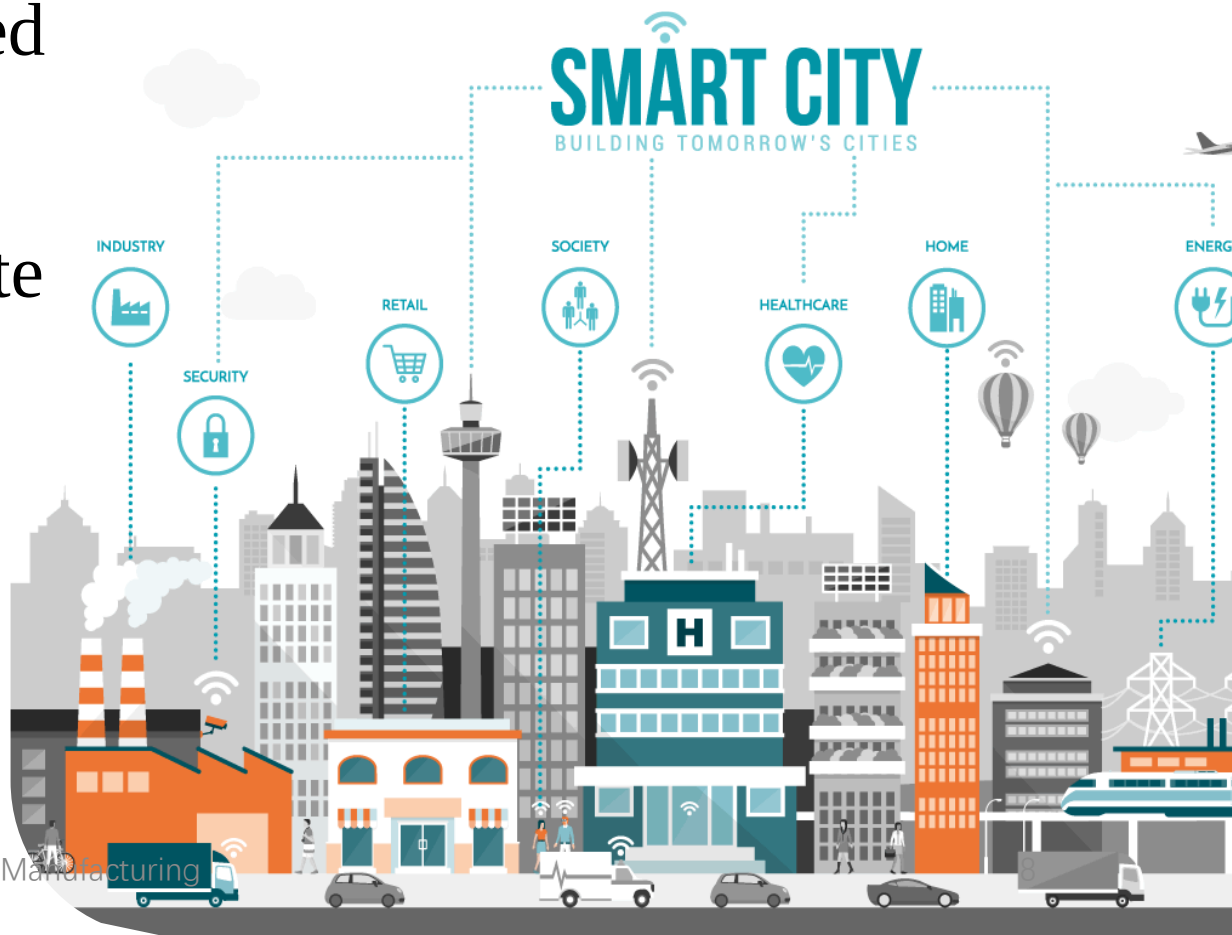


Intelligent Construction

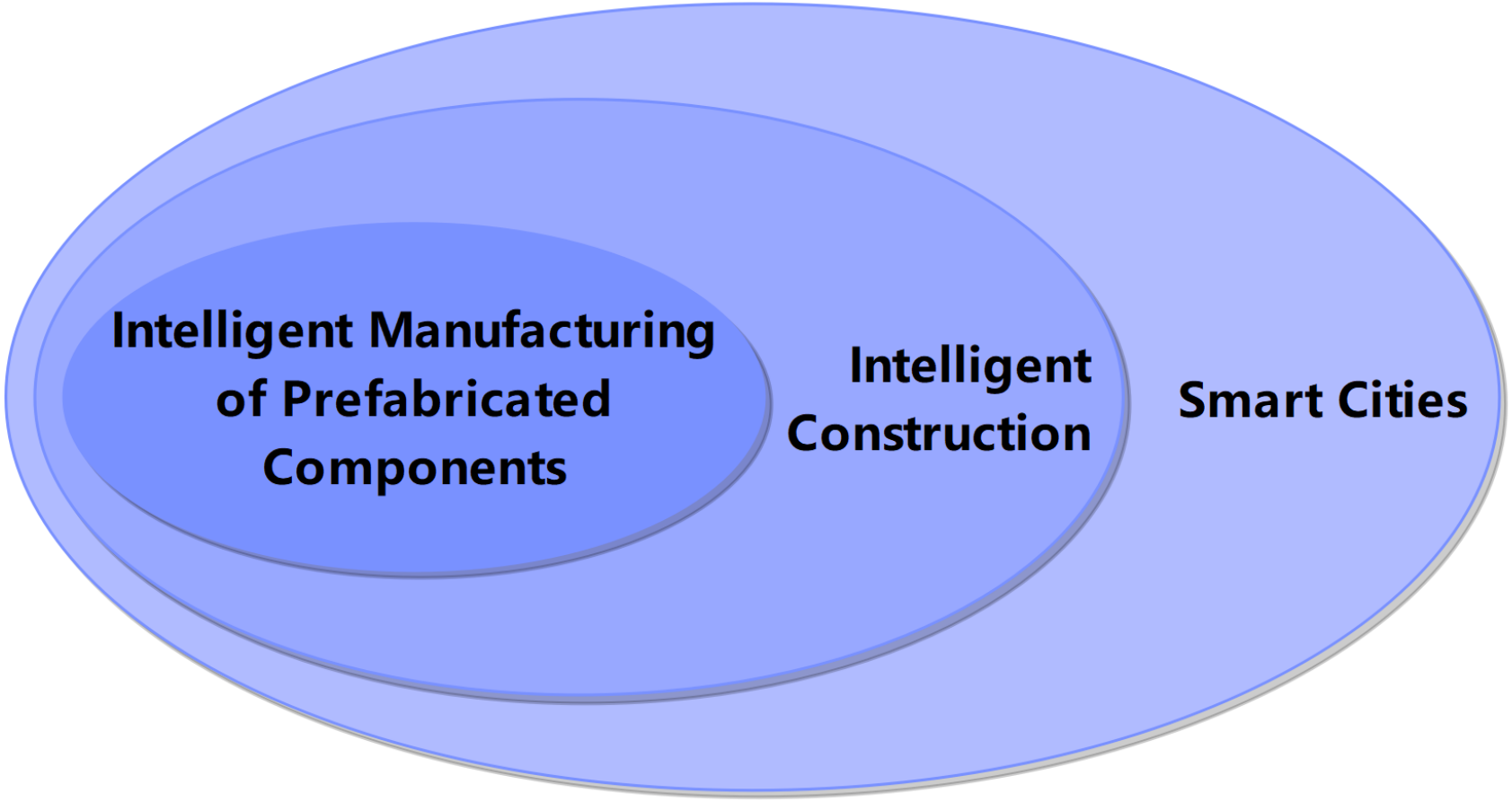
- **In the field of design:**
  - A component system, A more flexible structure.
  - Prefabricated buildings
- **In the field of construction:**
  - Purchasing materials→Producing components→Assembling into a building
  - FMS, RMS, Intelligent manufacturing technologies
- **In the field of operation and maintenance:**
  - BIM, IoT, big data, cloud infrastructure
  - “The integration of intelligent applications” & “The entry of big data”

Smart Cities

- **Geographic digital twins** formed by data from urban embedded sensors and API services have been popularized in **urban planning** practice, given the increasing appetite for digital technology in the Smart Cities movement.

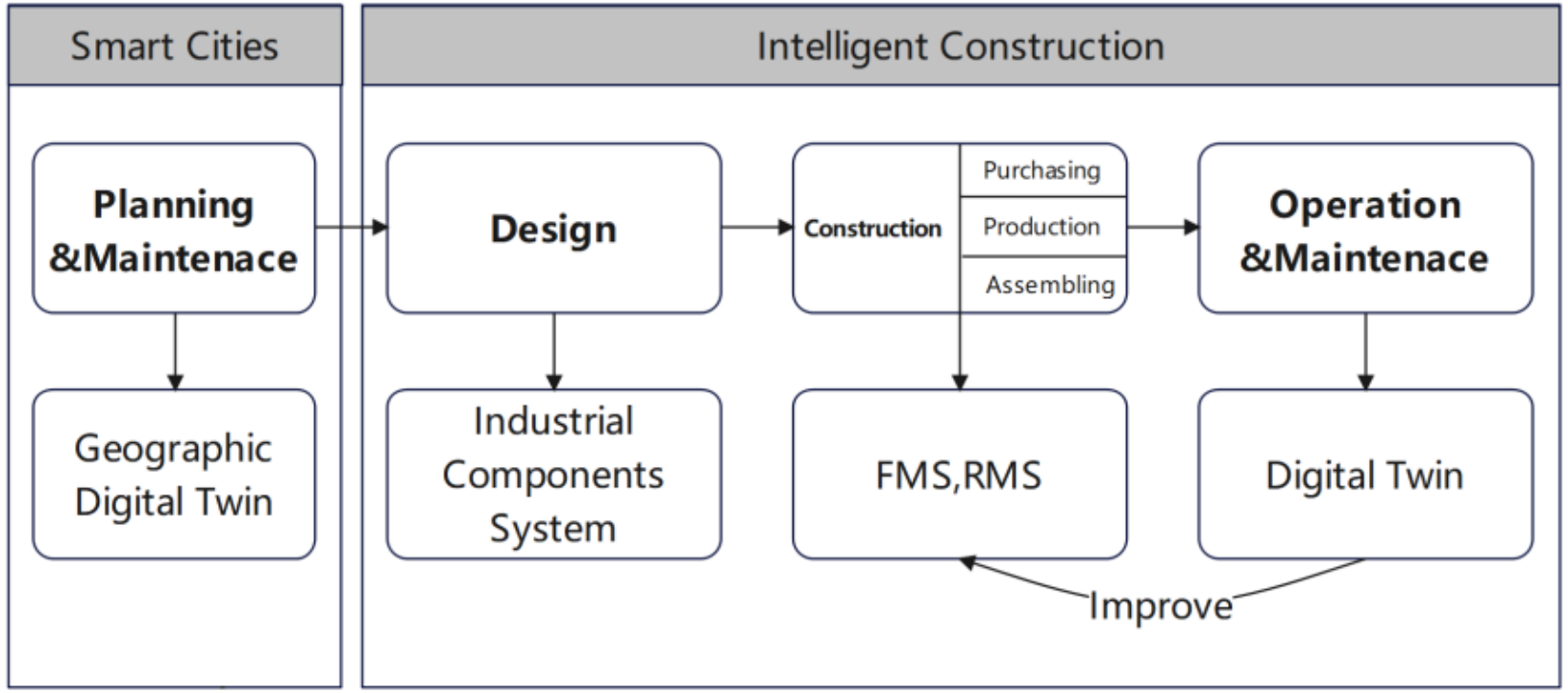


Conclusions



Conclusions

The intelligent Manufacturing shows its great potential and inspires the development of various industries in the context of industry 4.0.



THANKS!

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